

(TR-104) PYTHON WITH AI/ML

Training Day 13 Report

Date: 28 January 2026

Topic Covered: Pandas Library Basics, Data Analysis & Matplotlib Visualization

The thirteenth day of training focused on learning and applying Pandas for data manipulation and analysis along with basic data visualization using Matplotlib. The session included working on DataFrames, handling missing data, performing filtering operations, and creating basic charts for data representation.

- Created a DataFrame for employee records including Name, Department, Salary, and Years of Experience.
- Performed column selection and calculated average salary using `.mean()` method.
- Applied conditional filtering to extract employees based on experience and department.
- Updated salary values for employees in the Software Development department by 10%.
- Handled missing data using `.isnull()`, `.fillna()`, and `.dropna()` methods.
- Performed statistical analysis on dataset using `.describe()` method.
- Sorted data based on Clicks to identify top-performing records.
- Created a new feature column `Engagement_Score` for analysis purposes.
- Explored basic concepts of Matplotlib for data visualization.
- Created a bar chart to compare project counts across services.
- Created a line plot to visualize weekly website traffic trends with labels, title, and styling.
- **GitHub Link for the Tasks:**
https://github.com/JaspinderKaurWalia26/Python_Practice_Codes/tree/main/28-01-2026
- **Blog Link:** <https://medium.com/@jaspinderkaurwalia855/numpy-vs-pandas-4e080a7d4339>

Overall, the session helped in strengthening understanding of data manipulation using Pandas and basic data visualization techniques using Matplotlib for real-world analytical tasks.